IN THE HIGH COURT OF JUSTICE
BUSINESS AND PROPERTY COURTS OF ENGLAND AND WALES
TECHNOLOGY AND CONSTRUCTION COURT (KBD)
Claim No. HT-2026-000142

B E T W E E N:

MAYBROOK AGGREGATES LIMITED

Claimant

- and -

HELMSWORTH PLANT HIRE LIMITED

Defendant

EXPERT REPORT OF MR DAVID PENRITH

Prepared on the instructions of the Claimant pursuant to CPR Part 35

1. Introduction and instructions

1.1 I am David Penrith, BEng (Hons) CEng MIMechE, principal consultant of Penrith Crushing & Screening Consultants Ltd. I have 28 years' experience in the design, commissioning and failure investigation of mobile and static crushing and screening plant. My CV is at Appendix A.

1.2 I am instructed by Whitfield Carter LLP on behalf of Maybrook Aggregates Limited to investigate the causes of four failures of a HelmCrush HC-450 mobile crushing and screening plant (serial no. HC450-2231) supplied by Helmsworth Plant Hire Limited, and to address whether the Plant was capable of a sustained throughput of 320 tonnes per hour.

1.3 I understand that my duty is to the Court and not to those instructing me. I have complied with that duty and confirm that this report is my independent and impartial opinion. I have examined the Plant and the replaced jaw-crusher and screen components, the maintenance and breakdown records, the operating manual, the quarry feed and weighbridge records, and the parties' statements of case.

2. The Plant and its duty

2.1 The HC-450 is a tracked mobile plant combining a single-toggle jaw crusher (1100×700mm) with a triple-deck 5.4m screen. The maximum feed size specified in the operating manual is 650mm. The duty required by Maybrook was continuous primary and secondary crushing of quarried limestone at the Aldforth site.

2.2 I have reviewed the manufacturer's general arrangement drawings and the bearing schedule. In my opinion the toggle bearing (a single-row spherical roller bearing) and the jaw main-shaft bearing arrangement are under-specified for continuous quarry duty at the rated throughput. The dynamic load rating provides a calculated L10 bearing life of approximately 4,800 hours at 320 t/h, materially below the 24,000–30,000 hours expected for quarry-duty plant of this class.

3. The failures

3.1 **22 April 2025 – toggle bearing.** Examination of the failed bearing shows classic sub-surface fatigue spalling consistent with overload over time, not the brinelling or fracture that would indicate impact from tramp metal or oversized feed. There is no evidence of foreign-object ingress.

3.2 **19 May and 24 June 2025 – screen deck and drive bearing.** The cracked top deck is manufactured from 8mm plate. In my opinion 12mm plate is the appropriate minimum for limestone quarry duty; the deck failed by fatigue cracking initiating at the feed-end welds. The drive bearing seizure followed loss of lubricant through a failed seal of inadequate specification.

3.3 **28 July – 21 August 2025 – main-shaft bearing.** The main-shaft bearing failed catastrophically, damaging the crusher frame. The fracture surfaces and the progressive nature of the damage are consistent with the same under-specification and with degradation following the earlier failures, and not with a single misuse event. I found no metallurgical evidence of tramp-metal impact.

4. The allegation of misuse

4.1 I have considered Helmsworth's allegation that the failures were caused by oversized and contaminated feed and by ignoring warning alarms. Maybrook operated a primary scalping screen and a belt magnet upstream of the Plant. The quarry feed records and the photographs of the stockpiles are consistent with feed within the 650mm specification.

4.2 Had oversized boulders or tramp metal been fed to the crusher, I would expect to see characteristic impact damage to the jaw plates and brinelling of the bearings. I found neither. The breakdown log records that the Plant was stopped when alarms activated. In my opinion the misuse allegation is not supported by the physical evidence.

5. Throughput

5.1 I assessed throughput from the weighbridge records over the periods the Plant was operational. The sustained throughput achieved was approximately 200–215 tonnes per hour. In my opinion the Plant, even when operating correctly and fed in specification, was not capable of a sustained 320 tonnes per hour on this material; a realistic sustained figure is around 210 tonnes per hour. The 320 t/h figure appears to be a peak or test-condition figure that is not achievable in continuous quarry service.

6. Conclusions

6.1 The four failures were caused by inherent defects in the design and specification of the Plant (under-specified bearings and screen deck), not by misuse by Maybrook.

6.2 The Plant was not, when properly operated, capable of a sustained throughput of 320 tonnes per hour; its realistic sustained throughput on this material was approximately 210 tonnes per hour.

6.3 Helmsworth's response times to the reported breakdowns (averaging some 5 working days) were not consistent with good industry practice for plant under a maintenance contract.

Statement of compliance and truth. I confirm that I have complied with my duty to the Court under CPR Part 35 and PD35. I confirm that I have made clear which facts and matters referred to in this report are within my own knowledge and which are not. I believe the facts I have stated in this report are true and the opinions I have expressed are correct.

D. Penrith CEng MIMechE

Penrith Crushing & Screening Consultants Ltd – 14 April 2026